

Multiple Choice (10 marks)

1. Which of the following is NOT a property of bitmap graphics?
 - (a) Fast hardware exists to move blocks of pixels efficiently.
 - (b) Realistic lighting and shading can be done.
 - (c) All line segments can be displayed as straight.
 - (d) Polygons can be filled with solid colors and textures.
2. Which of the following statements is FALSE about memory reclamation based on reference counting?
 - (a) Reference counting is well suited for reclaiming cyclic structures.
 - (b) Reference counting incurs additional space overhead for each memory cell.
 - (c) Reference counting is an alternative to mark-and-sweep garbage collection.
 - (d) Reference counting need not keep track of which cells point to other cells.
3. Which of the following is a Wireless traffic Sniffing tool?
 - (a) Maltego
 - (b) BurpSuit
 - (c) Nessus
 - (d) Wireshark
4. Which of the following instruction-set features is NOT generally considered an obstacle to aggressive pipelining of an integer unit?
 - (a) Condition codes set by every instruction
 - (b) Variable-length encoding of instructions
 - (c) Instructions requiring widely varying numbers of cycles to execute
 - (d) Several different classes (sets) of registers
5. A hash function guarantees the integrity of a message. It guarantees that the message has not be
 - (a) Replaced
 - (b) Overview
 - (c) Changed
 - (d) Violated

True / False (10 marks)

Answer True or False

6. True or False: The types of scanning include network, vulnerability, and port scanning, which are essential for security assessments.
7. True or False: The maximum possible degree of the minimal-degree interpolating polynomial $p(x)$ for $n + 1$ distinct points is $n + 1$.
8. True or False: The number of distinct functions mapping a finite set A with m elements into a finite set B with n elements is $n!/(n - m)!$
9. True or False: A digital signature can be effectively created using a shared-key system without the need for public-key infrastructure.
10. True or False: Troya is classified as a remote Trojan that can facilitate unauthorized access to a user's system.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract every first or specified element from a given two-dimensional list.
12. Write a python function to count the total set bits from 1 to n .

End of Paper